

Gen-Sign: HLS based Watermarking using IP Vendor's Feistel Cipher Encrypted Genomic Signature for Protecting CNN and Image Processing Filter Cores against Piracy

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INTRODUCTION

- A CNN is a deep neural network widely used for visual tasks like image recognition, classification, and medical analysis, excelling at extracting patterns but requiring heavy computation.
- Image processing involves techniques for enhancing, altering, and analyzing images to achieve specific results.
- Key operations include sharpening filters for edge enhancement. Both CNN-based feature extraction and sharpening are computationally intensive, making dedicated hardware modules beneficial.
- These CNN accelerators and image filters can be designed as dedicated reusable intellectual property (IP) core using high level synthesis (HLS) framework.

INTRODUCTION

- The global design supply chain process makes the design process vulnerable to hardware threats of IP piracy and false claim of IP ownership.
- Therefore, securing IP designs from piracy is crucial to prevent revenue loss, reputational damage, and security risks from potential backdoor malicious logic.
- The proposed approach presents a security aware HLS watermarking methodology using IP vendor's Feistel cipher encrypted genomic signature for protecting CNN and image processing filter IP cores against piracy.



Fig. 1 : Hardware (IC) design process

PREVIOUS WORKS

Sr. No.	Existing Work	Technique Used	Remarks
1.	F. Koushanfar et. al., [12] (2005)	uses CIG based watermarking techniques	Generation of weaker watermark strength and becomes vulnerable in case of compromised digit encoding rule and signature size.
2.	D. Roy et al., [4] (2005)	exploited multilevel watermarking scheme	Secures IP design at both HLS and register transfer level. However, depicts weaker security due to limited watermarking strength
3.	J. Chen et. al., [8] (2021)	Pragma directive based HLA watermarking approach	uses synthesis directives to assign operations to functional units in the source code

PROPOSED WORK

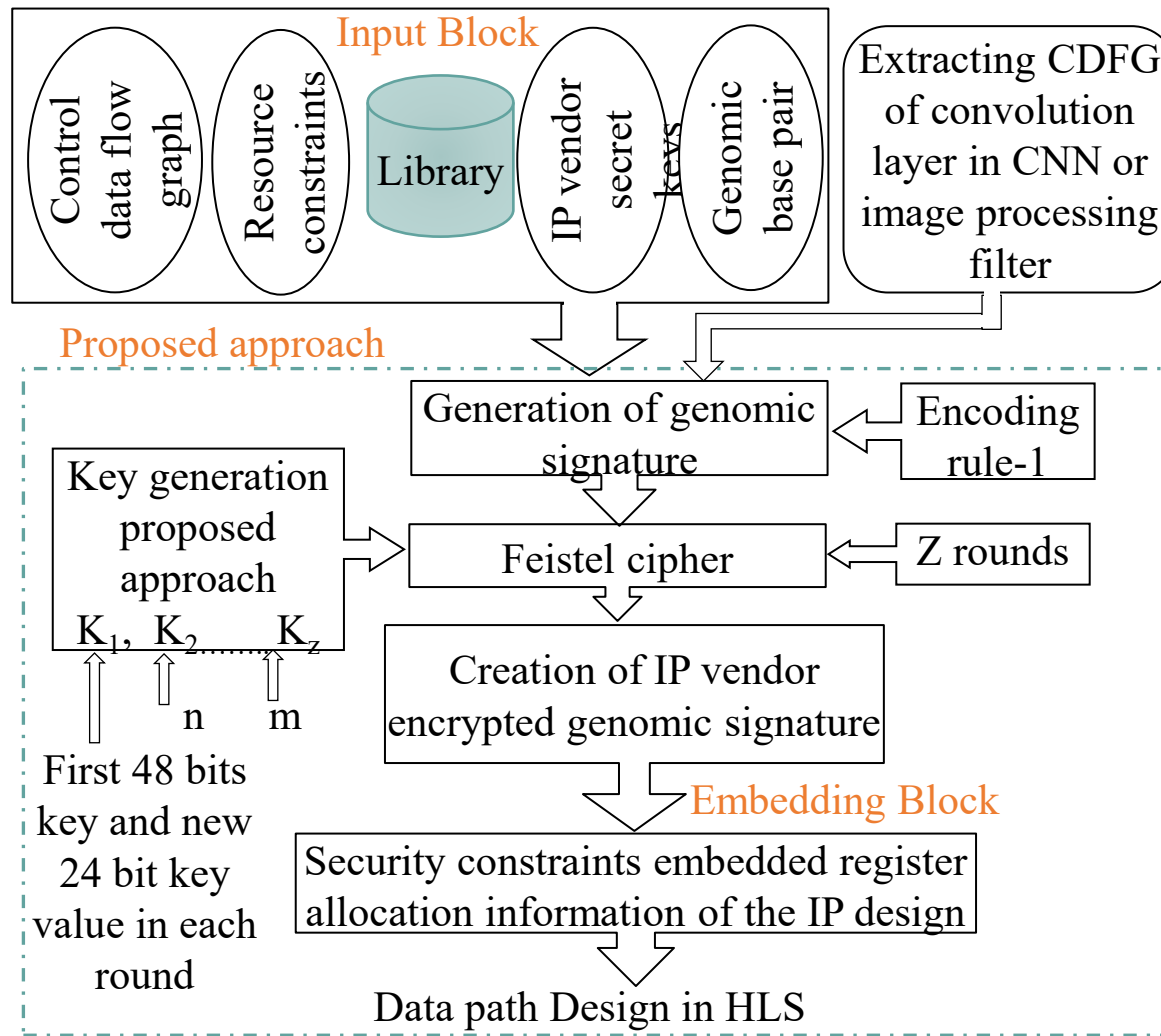
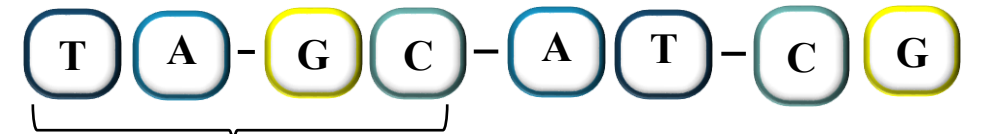


Fig. 1. Overview of Proposed Methodology (Gen-Sign)

Extraction of Genomic Sequence and Creation of Genomic Signature



Distinct Type

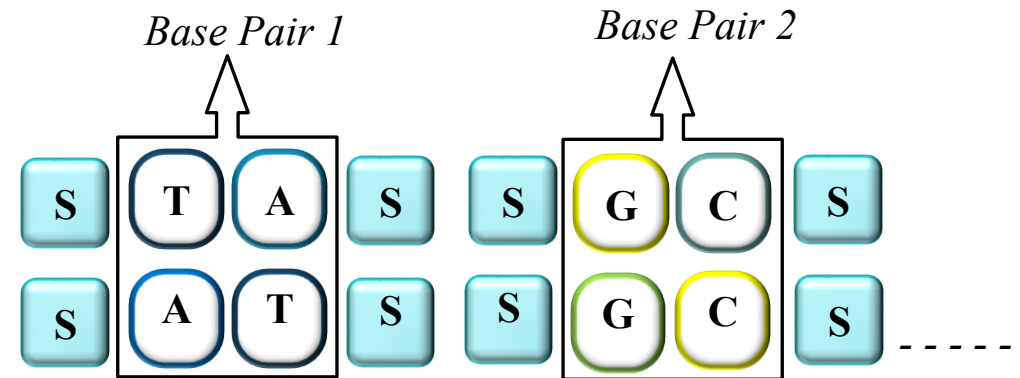
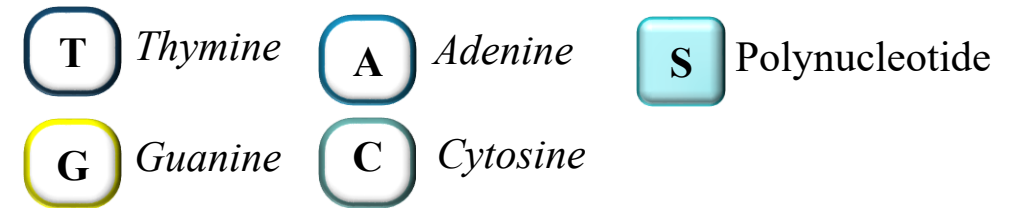


Fig. 2. Genomic sequence with base pairs and polynucleotide

PROPOSED WORK CONTINUE...

Generation of Encrypted Genomic Signature

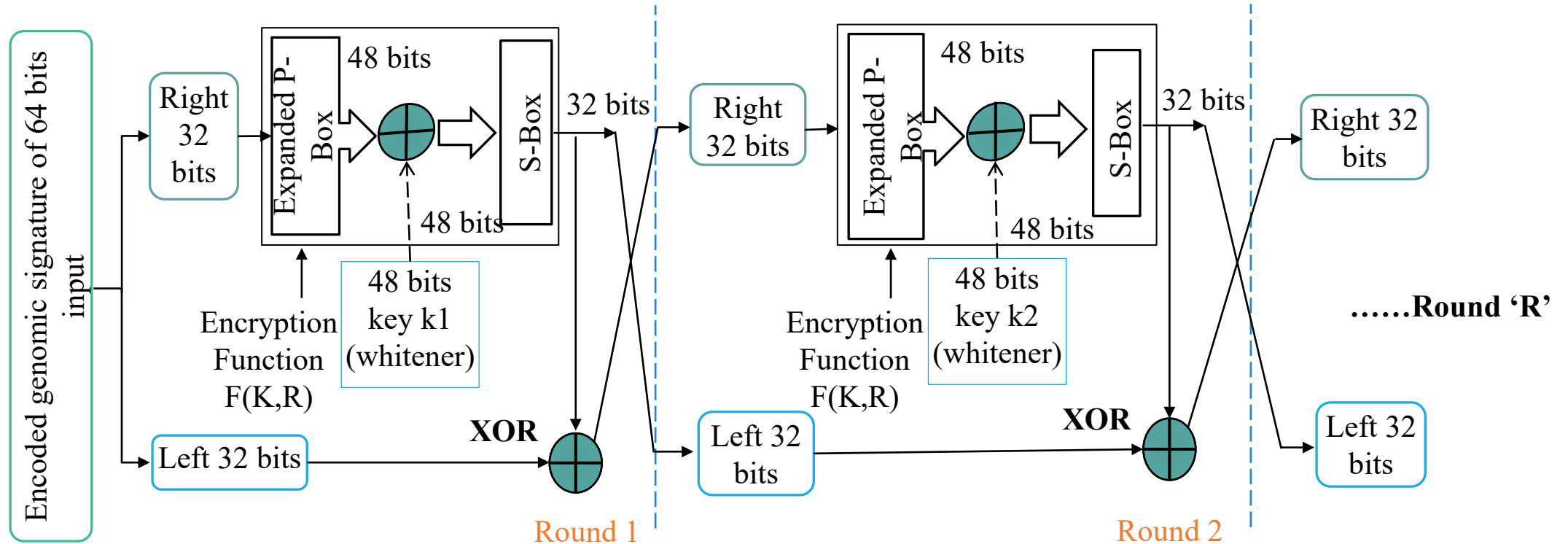


Fig. 3. Feistel cipher integrated in proposed Gen-Sign

PROPOSED WORK: CNN SDFG

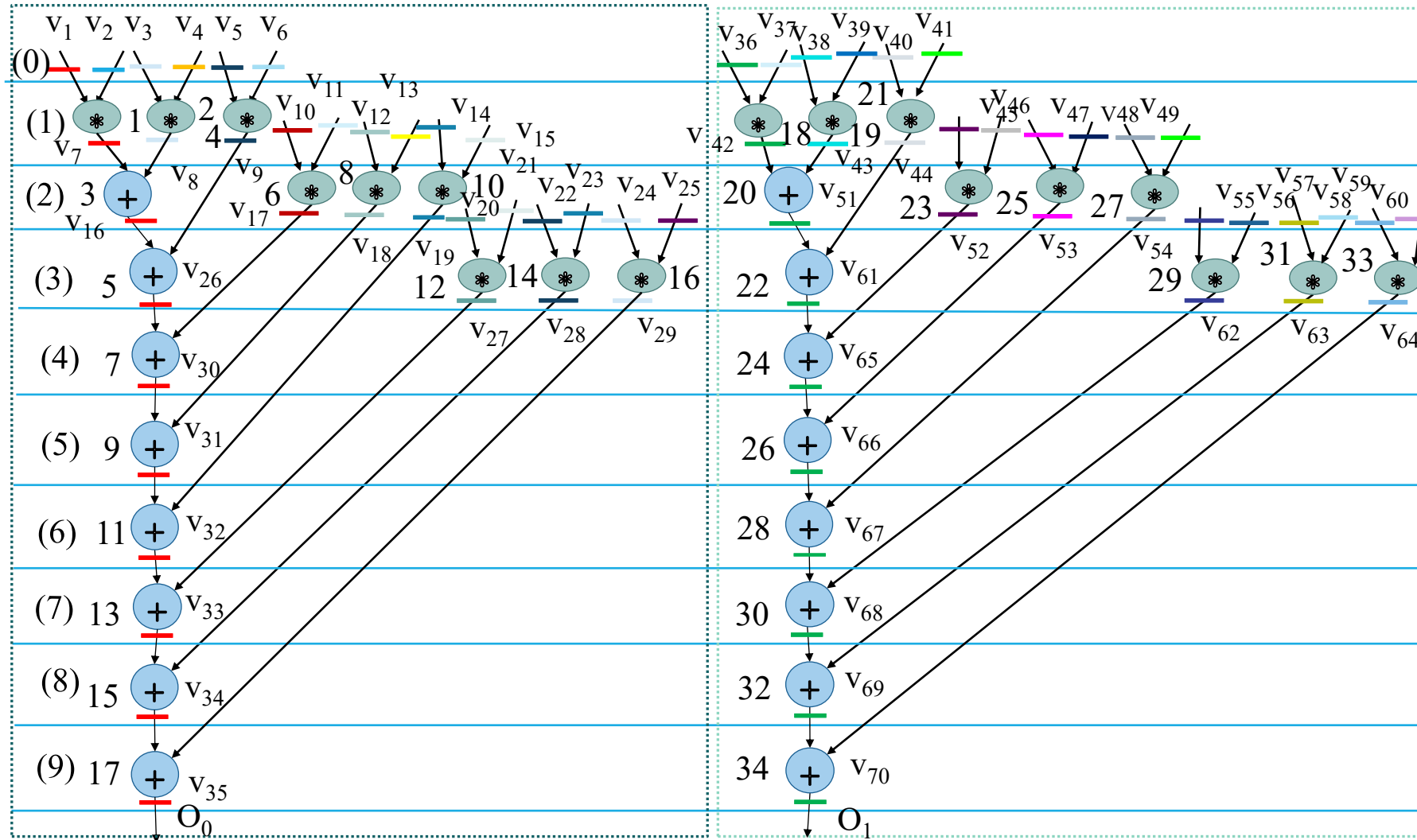


Fig. 4. Scheduled data flow graph of loop unrolled CNN using 6 M and 2 A

PROPOSED WORK: Sharpening filter SDFG

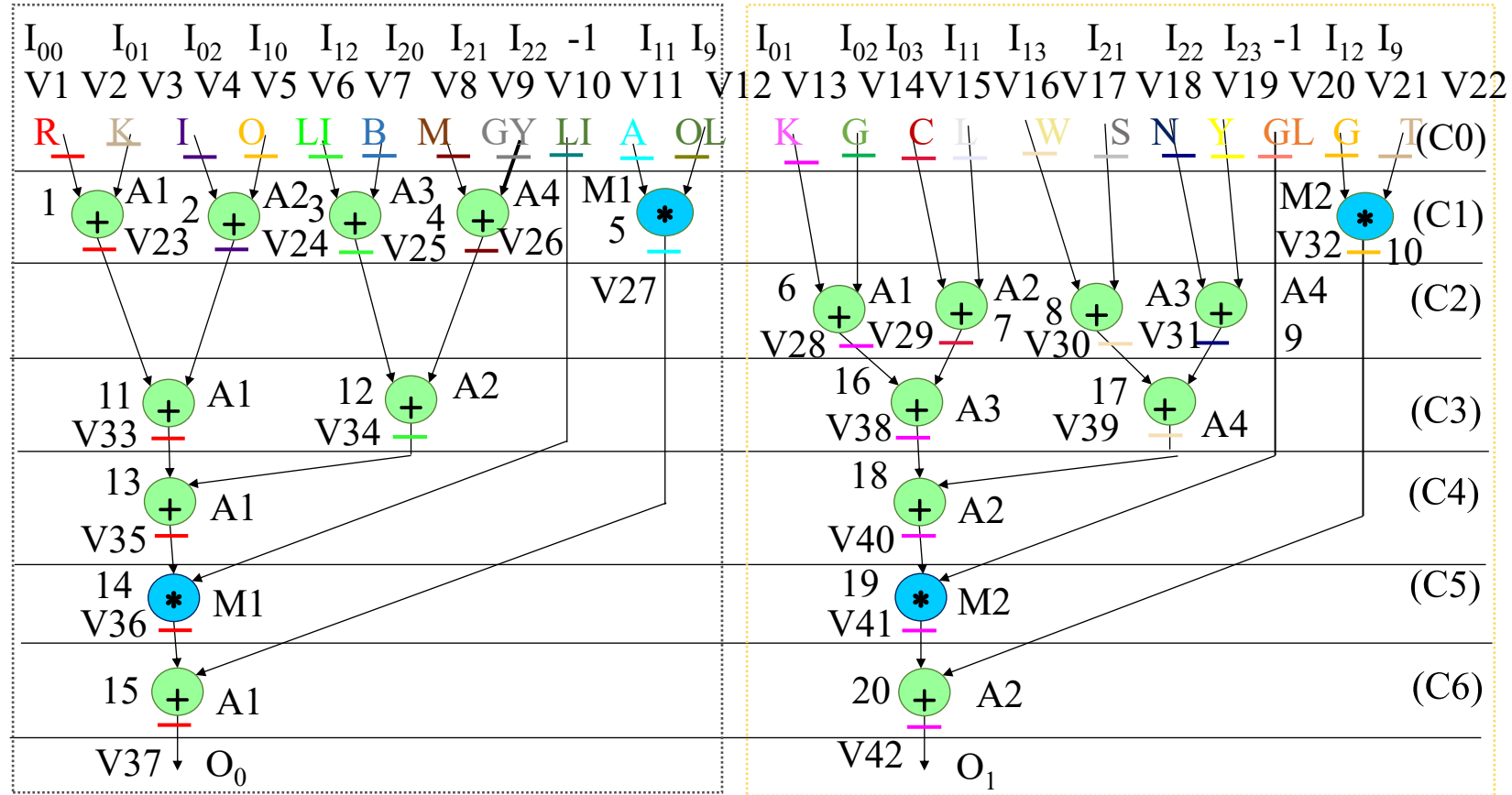


Fig. 5. Scheduled data flow graph for sharpening filter using 2M and 4A

PROPOSED WORK: Security Constraints and CNN RAT

TABLE 1. SECURITY CONSTRAINTS FOR CNN AND SHARPENING FILTER

CNN convolutional layer IP														
'0'bits	(v ₂ ,v ₄)	(v ₂ ,v ₆)	(v ₂ ,v ₈)	(v ₂ ,v ₁₀)	(v ₂ ,v ₁₂)	(v ₂ ,v ₁₄)	(v ₂ ,v ₁₆)		(v ₄ ,v ₅₂)	(v ₄ ,v ₅₄)	(v ₄ ,v ₅₆)	(v ₄ ,v ₅₈)	(v ₄ ,v ₆₀)	(v ₄ ,v ₆₂)
'1'bits	(v ₁ ,v ₃)	(v ₁ ,v ₅)	(v ₁ ,v ₇)	(v ₁ ,v ₉)	(v ₁ ,v ₁₁)	(v ₁ ,v ₁₃)	(v ₁ ,v ₁₅)		(v ₃ ,v ₅₇)	(v ₃ ,v ₅₉)	(v ₃ ,v ₆₁)	(v ₃ ,v ₆₃)	(v ₃ ,v ₆₅)	(v ₃ ,v ₆₇)
Sharpening Filter IP														
'0'bits	(v ₂ ,v ₄)	(v ₂ ,v ₆)	(v ₂ ,v ₈)	(v ₂ ,v ₁₀)	(v ₂ ,v ₁₂)	(v ₂ ,v ₁₄)	(v ₂ ,v ₁₆)		(v ₈ ,v ₁₆)	(v ₈ ,v ₁₈)	(v ₈ ,v ₂₀)	(v ₈ ,v ₁₆)	(v ₈ ,v ₁₈)	(v ₈ ,v ₂₀)
'1'bits	(v ₁ ,v ₃)	(v ₁ ,v ₅)	(v ₁ ,v ₇)	(v ₁ ,v ₉)	(v ₁ ,v ₁₁)	(v ₁ ,v ₁₃)	(v ₁ ,v ₁₅)		(v ₇ ,v ₁₅)	(v ₇ ,v ₁₇)	(v ₇ ,v ₁₉)	(v ₇ ,v ₂₁)	(v ₇ ,v ₂₃)	(v ₇ ,v ₂₅)

TABLE 2. RAT PRE AND POST EMBEDDING WATERMARK FOR CNN

[illegible]

PROPOSED WORK: Sharpening Filter RAT

TABLE 3. RAT BEFORE AND AFTER EMBEDDING HARDWARE
SECURITY CONSTRAINTS CORRESPONDING TO SHARPENING FILTER

Registers	C0	C1	C2	C3	C4	C5	C6
Red(R)	V1	V23	V23	V33	V35	V36	V37
Khaki(K)	V2	V23	V23	V33	V35		V37
Indigo(I)	V3	V24	V24				
Orange(O)	V4						
Lime(LI)	V5	V25	V25	V34			
Blue(B)	V6	V25	V25				
Maroon(M)	V7	V26	V26				
Gray(GY)	V8						
Teal(T)	V9	V9	V9	V9	V9		
Aqua(A)	V10	V27	V27	V27	V27	V27	
Olive(OL)	V11						
Purple(P)	V12	V12	V28	V38	V40	V41	V42
Green(G)	V13	V13					
Crimson(C)	V14	V14	V29				
Lavender(L)	V15	V15					
Wheat(W)	V16	V16	V30	V39			
Silver(S)	V17	V17					
Navy(N)	V18	V18	V31				
Yellow(Y)	V19	V19					
Salmon(S)	V20	V20	V20	V20	V20		
Gold(GL)	V21	V32	V32	V32	V32	V32	
Tan(T)	V22						

RESULT AND ANALYSIS

➤ Tamper Tolerance :

$$TT = S^a$$

Where, S and a corresponds to types of watermark signature bits and generated watermarking strength of the corresponding security approach.

➤ Probability of Coincidence :

$$PC = \left(1 - \frac{1}{r}\right)^s$$

Where, the variable 'r' stands for the total number of registers present before the integration of hardware security constraints. Meanwhile, 's' symbolizes the total count of hardware security constraints that have been embedded into the system

➤ Design Cost :

$$Cf(Fi) = W_1 \left(\frac{L_d}{L_m}\right) + W_2 \left(\frac{A_d}{A_m}\right)$$

Where L_d and A_d represent the design latency and area respectively, and L_m and A_m denote the maximum latency and area. W_1 and W_2 are the designer-specified weights, set to 0.5 to ensure equal priority between the factors

RESULT AND ANALYSIS

TABLE 4. OVERHEAD ANALYSIS OF PROPOSED GEN-SIGN

Bench- mark	Regis- ter Count	Baseline Design			Embedded Design			Design Cost Over head
		Area (μm^2)	Latency (ps)	Cost	Area (μm^2)	Latency (ps)	Cost	
CNN	36	625.11	1192.36	0.33	625.11	1192.36	0.33	0%
Sharpen- ing Filter	22	243.62	794.9	0.58	243.62	794.9	0.58	0%
Blur Filter	20	109.96	1523.54	0.83	109.96	1523.54	0.83	0%
LED Filter	14	199.64	728.66	0.68	199.64	728.66	0.68	0%

TABLE 5. COMPARISON OF TT WITH
PREVIOUS WORKS

Benchmarks	CNN	Sharpe ning Filter	Blur Filter	LED Filter
Proposed <i>Gen-Sign</i>	3.40 E+38	3.40 E+38	3.40 E+38	3.40 E+38
Physical-Level Watermarking [3]	256	256	256	256
Multilevel Watermark [4]	16384	16384	16384	16384
Facial Biometric [5]	9.67 E+24	9.67E+2 4	9.67 E+24	9.67 E+24
Automated Watermark Methodology [6]	1.15 E+18	1.15 E+18	1.15 E+18	1.15 E+18

RESULT AND ANALYSIS

TABLE 6. COMPARISON OF PC WITH PREVIOUS WORKS

Benchmarks	CNN	Sharpen- ing Filter	Blur Filter	LED Filter
Register	36	22	20	14
Proposed <i>Gen-Sign</i>	2.71E-2	2.59E-3	1.40E-3	7.59E-5
Physical-Level Watermarking [3]	7.98E-1	6.89E-1	6.63E-1	5.52E-1
Multilevel Watermark [4]	6.74E-1	5.21E-1	4.87E-1	3.54E-1
Facial Biometric [5]	9.65E-2	2.10E-2	1.41E-2	2.1E-3
Automated Watermark Methodology [6]	1.84E-1	6.13E-2	4.60E-2	1.17E-2

TABLE 7. COMPARISON OF ENTROPY WITH PREVIOUS WORKS

Hardware Security Approach	Entropy
Proposed Gen-Sign	1.04E-53
Physical-Level Watermarking [3]	3.9E-3
Multilevel Watermark [4]	6.10E-5
Facial Biometric [5]	2.59E-33
Automated Watermarking Methodology [6]	8.67E-19

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Thank You!